

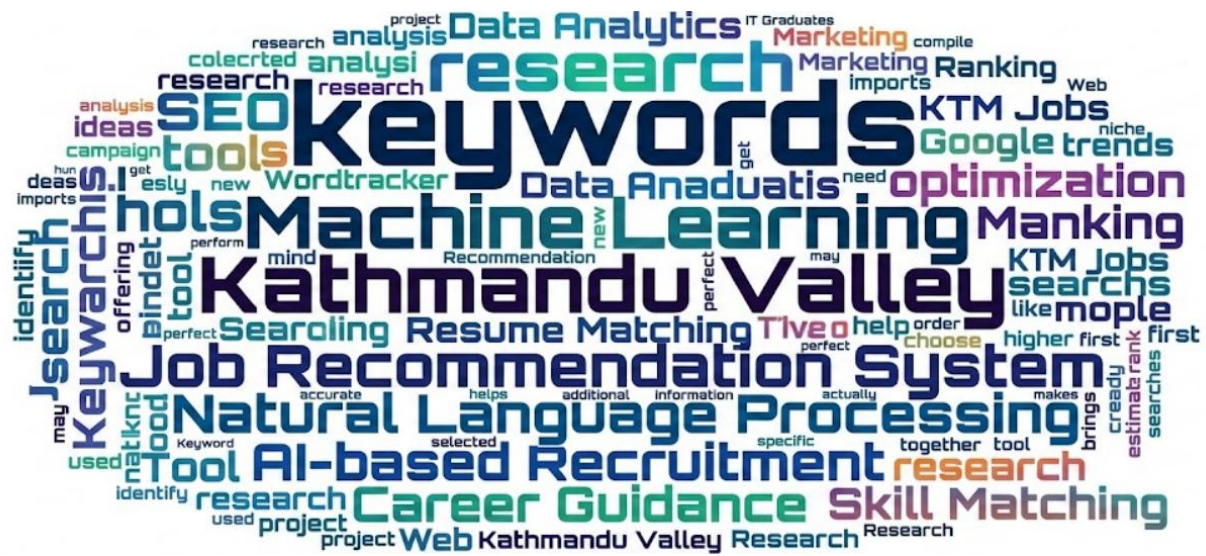
ST6000CEM Individual Project Preparation

Design and Development of a Machine Learning-Based Resume Matching and Job Recommendation System for IT Graduates in Kathmandu Valley



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Keywords



Contents

Introduction.....	5
Aim	6
Objectives	6
Statement of the Problem.....	7
Research Questions.....	7
Literature Review.....	8
Summary of Existing Research.....	8
Case Studies	8
Research.....	9
Methodology	10
Software Development Life Cycle.....	10
Compliance and Ethical Considerations	10
Flowchart / Block Diagram.....	11
Tools and Technologies.....	12
Integration	13
Project Plan	14
Potential Risks and Mitigation Strategy	15
Conclusion	16
References.....	17
Appendix.....	18

Table of figure

Figure 1 System architecture	11
Figure 2 Tech stack	12
Figure 3 System flow	13
Figure 4 Project plan	14
Figure 5 Swot anlaysis	18
Figure 6 Pestel Analysis.....	19

Introduction

As an IT student in Nepal, I have seen how challenging it can be for graduates to find jobs that fit their abilities they actually want to do. Every year, thousands of IT graduates (many from colleges within the Kathmandu Valley) who possess programming, database management, web development and even machine learning skill will apply for hundreds of job yet every year there seems to be an overwhelming number of graduates with no job prospects. Although there are hundreds of applications for IT positions, applicants receive interviews or eventually become employed.

I have spoken to many of my peers and have completed mine internships, and one commonality I have found is how companies evaluate resumes. Many companies still conduct their initial screening procedure manually or use simple keyword-based technology that does not accurately define the skills identified on the resume.

There are already many global job recruiting platforms that help individuals find jobs, such as LinkedIn, but none of them have the ability to reflect local context, especially outside of a major metropolitan area. Therefore, the intent of my project was to develop a machine learning-based system to read resume extract the proper skill set and match the IT opening within Nepal. Keeping this in mind was Essential in guiding my decision making process from developing a way to collect my data to selecting the proper machine learning algorithms to implement into the system.

Aim

The purpose of this project is to provide a better way for IT graduates to locate suitable employment opportunities. Rather than using only traditional keyword matching, I want the system to understand how well a candidate's background, education, and projects fit together. This should help eliminate frustration and wasted time for graduate and employee by assisting them with finding appropriate match between candidate and job opening. All aspects of the project including how data was collected, what algorithms are used, etc. were guided by this purpose; thus, ensuring that the resulting system is practical, relevant, and actually useful to people working in information technology within Nepal.

Objectives

To accomplish these goals, I created a number of concrete objectives. The first of these is to gather and compile a dataset of resume and job posting from IT company in Nepal. As part of this objective, I will also work to identify key skill, experience and qualification on resume in each in order to identify resumes to find the right people. Next, develop a Model to look at all resumes and job postings, and be able to accurately match candidates to those jobs through the use of a machine learning algorithm, this algorithm will be tested for quality and adjusted for equity and relevancy. Finally, I will develop user-friendly interface for both Graduates and Employers. This objective served as guideline throughout development and ultimately delivered a system that functions as intended while continuing to address the ongoing struggle of IT Graduates in Nepal.

Statement of the Problem

I am student from Kathmandu and I have personally observed the difficulty that recent IT graduates face when searching for job that truly match their skill. Many graduates complete their degree with strong knowledge in emerging areas. Despite this preparation, they often struggle to secure positions that reflect their actual capabilities. One major reason for this issue is the way resumes are seen. Many small and mid-sized IT companies rely on manual screening, which can be time consuming and inconsistent. Others use automated keyword-based systems that fail to understand context. For example: a candidate who has completed a Deep Learning project using a candidate with TensorFlow skill may be missed for a machine learning role if the keyword is not detected. does not match the job description even though the candidate has required skill.

This mismatch between skill and job requirement waste time for both applicant and employer and discourages young professional. Although global platforms like LinkedIn and Indeed offer support but they are not fully adapted to Nepal local IT market. Therefore, The aim of this project is to create a machine learning–based system that helps match resumes with suitable jobs in the local market.

Research Questions

How well can the system recognize a candidate skill and experience from their resume?

How effectively can it match IT graduate in Kathmandu Valley to job that fit their abilities and local market need?

What steps are needed to prevent bias ensure fairness to protect candidate data during the recommendation process?

How can feedback from graduate and employee be used to refine the system and improve recommendation accuracy over time?

Literature Review

Summary of Existing Research

Global resume matching platform such as LinkedIn, indeed, and Glassdoor, have traditionally depend on keyword based filtering and common machine learning technique. These approach are efficient for managing large volume of data but often issue to understand the deeper context behind a candidate skill or experience. Recent studies emphasize the advantages of natural language processing (NLP) method such as named entity recognition, word embedding, and semantic similarity for capturing information from resume. Supervised machine learning models, including Decision Trees, Random Forest, and Support Vector Machines are frequently combined with collaborative filtering to improve candidate job alignment. Additionally, hybrid approach by integrating NLP with recommender system have shown promise in improving prediction accuracy. Visual analytics tool, such as clustering plot, heatmap, and skill distribution chart provide valuable insight into employment patters, highlighting mismatch, skill shortage and emerging demand areas (Sinha et al., 2023). These methods help not only in matching candidate but also in guiding strategic workforce planning.

Case Studies

Research focused particularly through observation and internship in Kathmandu Valley highlight, challenges overlooked by global platforms. Local resume often employs varied terminology to describe similar competency while employers place strong focus on real world experience and relevant specific skill rather than formal certification (Sharma et al., 2025). Consequently, datasets must be carefully curated and preprocessed to account for inconsistent or incomplete resume. Incorporating feedback from graduates and smaller IT firms proves essential for refining recommendation. Studies indicate that context aware, locally tailored systems are far more effective than generic models at representing the realities of the IT job market in Nepal. Moreover, small scale experiments demonstrate that integrating employer input during model training increase trust and adoption as stakeholder notice the system to be responsive to local hiring practices (Dangeti et al., 2026).

Research

Even with progress in NLP and machine learning, many gaps remain in current research and practice. Global platform rarely considers region specific resume convention or employer priorities (Brindashree & Pushphavathi, 2023). Keyword based method often overlook semantic relationship between skill and role, while visual analytics remain underutilized for local stakeholder limiting actionable insight. Ethical consideration, including fairness, bias mitigation, and data privacy are inconsistently addressed across current system. Furthermore, there is a lack of longitudinal study examining how machine learning based recommendation affect career outcomes over time. These gaps suggest a pressing need for a locally relevant context sensitive system that integrates NLP supervised learning, iterative feedback and ethical safeguard. The system could assist IT graduates in Nepal in finding fitting jobs and give employers useful insight (Ertuğrul & Bitirim, 2025).

Methodology

Software Development Life Cycle

Requirement analysis determines functional requirements, such as correctly identifying skills from resume, reliable candidate job matching, privacy of candidate data, and an intuitive interface. Nonfunctional requirements include scalability, efficiency, and maintainability. Local survey interview and observation ensure the system reflects Nepalese IT employment reality.

System Design adopts a modular architecture integrating NLP and ML workflow, enabling resume parsing, skill extraction, and recommendation generation. Resume Preprocessing handles local terminology, formatting inconsistencies, and missing information to prepare clean datasets. Machine Learning Model Development employs supervised models like Random Forest and Gradient Boosted Trees, evaluated using precision, recall, and F1 score with iterative retraining for evolving trends.

The system integrates content-based and collaborative filtering with a backend handling data and machine learning tasks and a frontend offering registration, resume upload and job search features. Testing and Iteration involve unit testing, integration testing and user acceptance testing guided by continuous feedback.

Compliance and Ethical Considerations

Data privacy and security are central to the system. Candidate and employer information is protected through encryption, secure authentication and role-based access control with regular audits ensuring authorized access only. Bias mitigation strategy is applied to machine learning models to ensure fairness across gender, ethnicity, and educational background while transparent logging allows stakeholders to understand recommendation decisions. Ethical data usage is maintained through consent-based collection and clear policy for data correction or removal. The system uses secure cloud hosting, routine backups and disaster recovery measures to ensure reliable operations. Continuous feedback from users informs both system improvement and ethical oversight ensuring recommendations remain fair, transparent and responsible over time.

Flowchart / Block Diagram

Following an Agile workflow the system begins with candidate registering and uploading resumes, while employers post job. Resumes are processed using NLP to extract skill, education, and experience, then cleaned for local formatting inconsistency. Machine learning model train on this data to match candidate with jobs via a hybrid recommendation engine. Continuous feedback from candidate and employer inform iterative retraining and system refinement. The system enforces compliance and ethics using role-based access, strong encryption, and clear audit logs. The final output provides intuitive dashboards, ensuring accurate, fair and responsive job recommendation aligned with Agile principles.

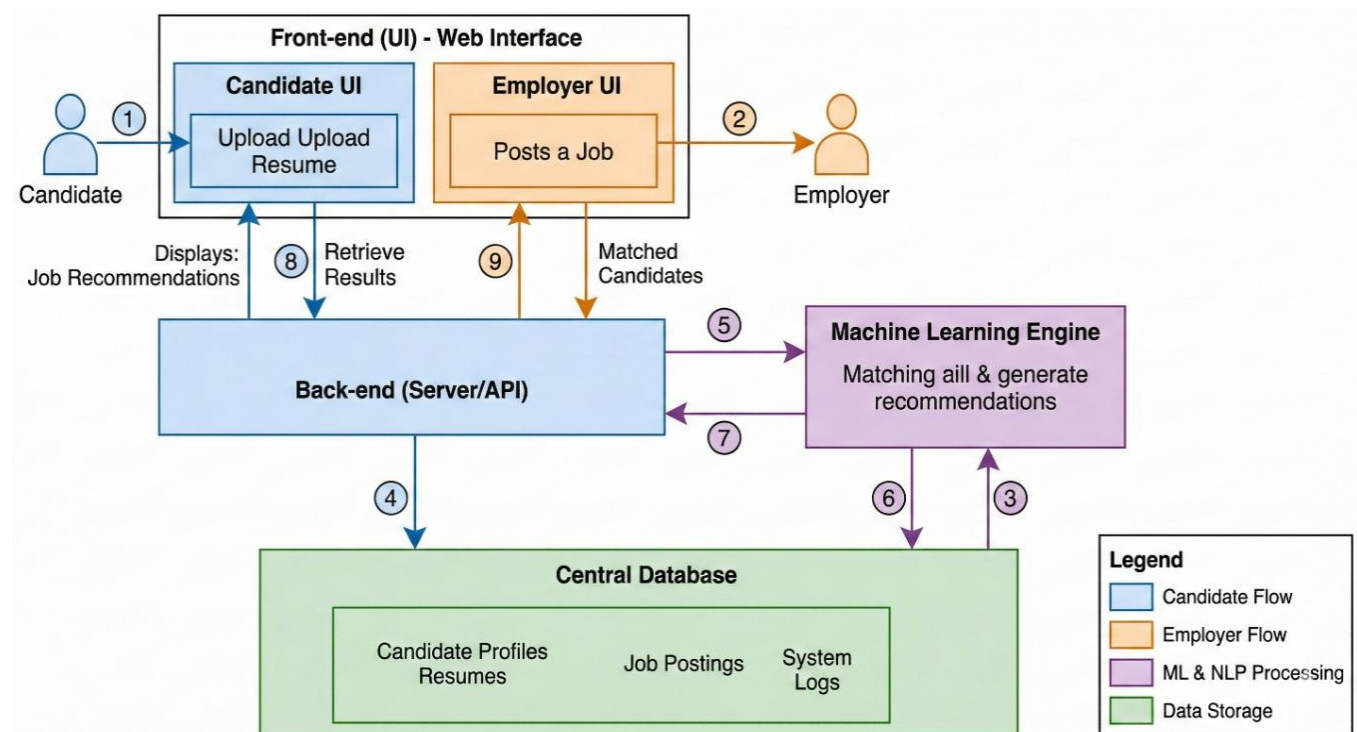


Figure 1 System architecture

Tools and Technologies

The tools selected for this project were chosen to balance technical capability, scalability and relevance to Kathmandu Valley IT graduate. Python serve as the primary programming language because of its robust ecosystem for machine learning and natural language processing.. Libraries such as scikit-learn support supervised learning models, while NLTK enable resume parsing, named entity recognition, and semantic similarity analysis. Pandas and NumPy assist with data cleaning, preprocessing and numerical operations forming a reliable foundation for accurate model training.

For the web platform, Django manages backend logic, integrating ML models with database operation, while ReactJS provides an interactive front end for candidates and employers to upload resume, search for jobs and view recommendations. PostgreSQL stores structured resume and job data efficiently supporting scalability as the system grow.

Git is used for version control and Docker provide consistent deployment environments strengthening development practice. Matplotlib and Seaborn produce visualization including heatmap and clustering plots helping to explore and communicate patterns in skills and job matches. Together, these tools enable a user centered context-aware system that delivers ethical, effective and locally meaningful job recommendations (Hull, 2024).

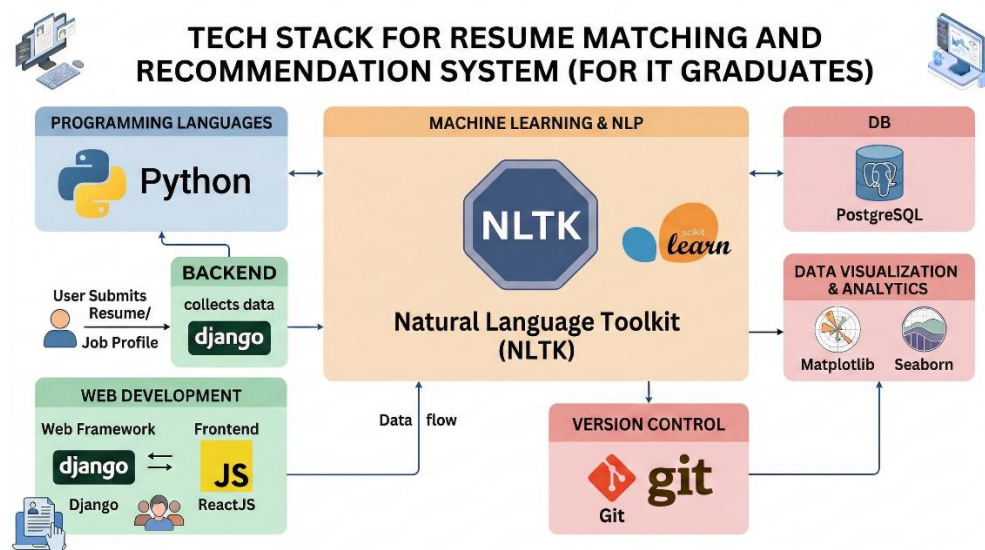


Figure 2 Tech stack

Integration

The system brings together candidates, employers and the matching engine. Candidates can upload resumes, manage profiles and get job recommendations based on their skills and experience while employers can post jobs and access a curated list of qualified candidates to streamline hiring.

The matching engine review resume, identifies key skill, and recommend jobs that fit both the candidate and the employer. A central database keeps all resume, job postings and record secure while ensuring smooth system operation and supporting future growth. Together, these components form a reliable, fair and user focused system that helps IT graduates in Kathmandu Valley find jobs that match their abilities and potential.



Figure 3 System flow

Project Plan

The project uses an Agile iterative approach to ensure continuous improvement at every stage. It begins by talking to Kathmandu IT graduates and small IT companies to understand real challenges in job matching. These insights shape the system design, including modular architecture, database structure and workflows for resume parsing, skill extraction and recommendation.

Next, the data phase cleans resume and job postings, addresses local terminology and builds a dataset ready for machine learning. Model development and refinement involve training supervised learning models validating them continuously and integrating them into the recommendation engine to ensure fair and accurate matches. The frontend and backend are developed iteratively, providing users with an intuitive and user friendly interface. Regular testing and feedback cycle refine usability, system performance and recommendation accuracy (Gordon, 2025). This plan balances flexibility, local context, and ML, ensuring a reliable, practical and effective system for Kathmandu Valley IT graduates.

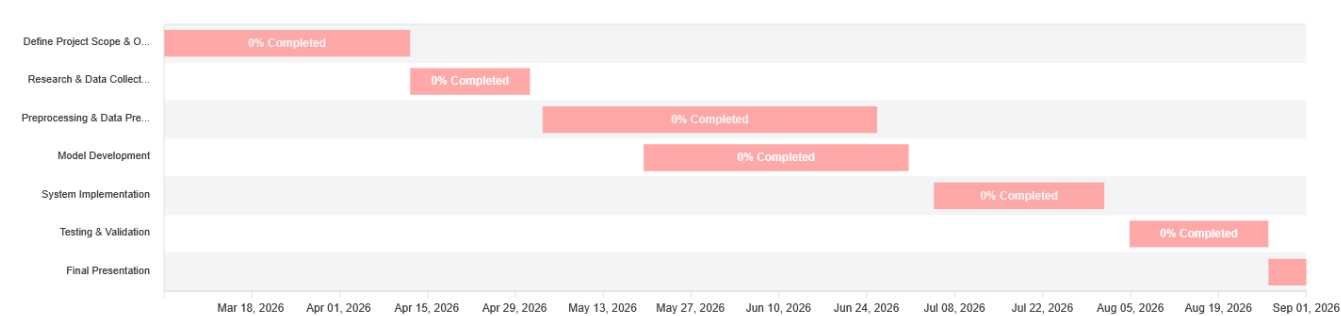


Figure 4 Project plan

Potential Risks and Mitigation Strategy

Risk	Description	Mitigation	Impact
Low-quality resumes	Resumes may have missing information or inconsistent formatting	Clean and standardize data by leverage NLP techniques	High
Limited local data	Few Nepalese resumes or job postings available	Collect data from universities and IT firms with augment datasets	Medium
Recommendation bias	System may favor certain skills or candidates	Apply fairness checks and validate recommendations using local data	High
Usability issues	Interface may be confusing for users	Conduct user testing and refine the design based on feedback	Medium

Conclusion

This project addresses a critical challenge in the Kathmandu Valley IT job market by developing a machine learning based system that matches resumes with suitable job postings for Nepali IT graduate. Its primary goal is to connect candidates with positions that align with their skills while helping companies streamline recruitment and reduce the time spent on manual resume screening. By leveraging natural language processing, the system goes beyond simple keyword matching to establish more accurate and meaningful connections between applicants and employers.

In addition to practical advantages, the project shows how technology can support ethical and transparent hiring in Nepal. The system improves recruitment efficiency, speeds up quality hiring and lays the groundwork for AI-assisted job placement initiatives. By connecting education with employment, it helps graduates apply their skills effectively, fostering a fair, data driven and efficient job market.

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Appendix

PotentialRisk:

https://docs.google.com/spreadsheets/d/1V1JkF5Ms8mRnbVQMqyFImiwu8oJctamEv7j7_hbPYP8/edit?usp=sharing

SWOT Analysis

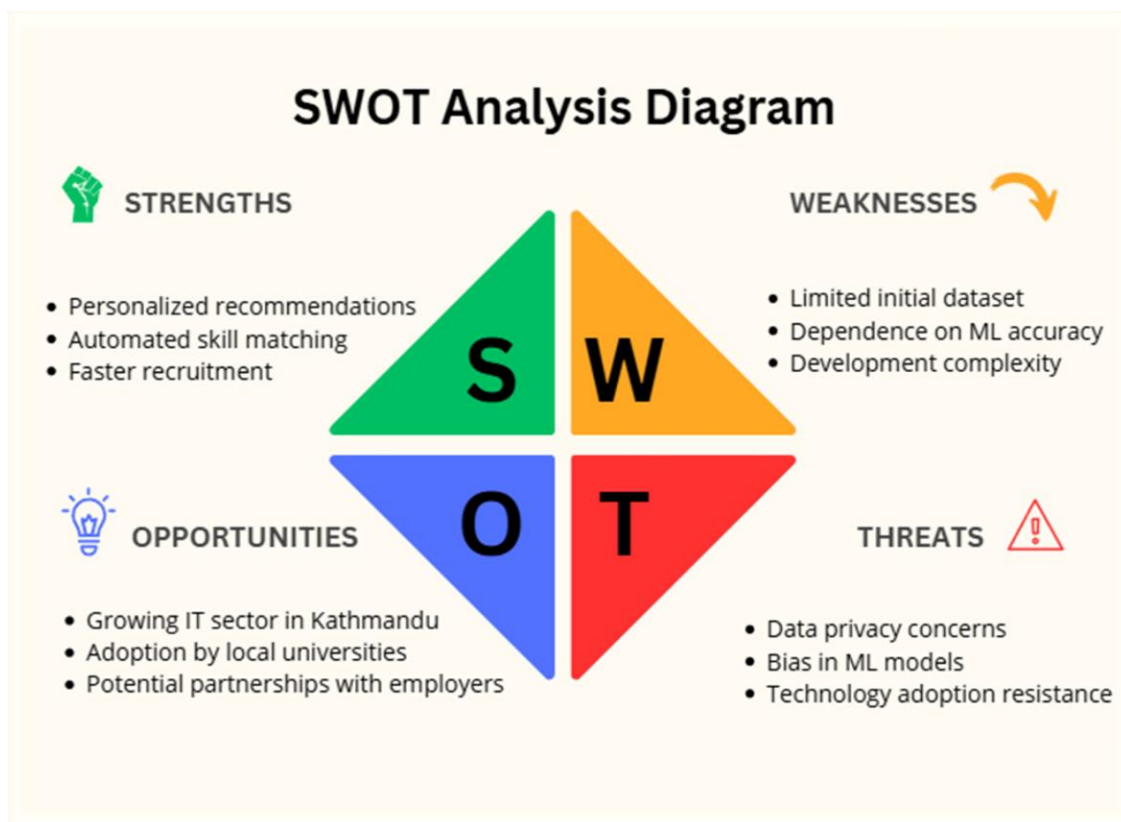


Figure 5 Swot anlysis

PESTEL Analysis

PESTEL ANALYSIS



Figure 6 Pestel Analysis